IDEOLOGICAL POLARIZATION IN CLIMATE CHANGE DISCOURSE

An NLP analysis of YouTube political commentators

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Research Question

How does political ideology shape the language and framing of climate change discussions on YouTube?

Can linguistic patterns reveal whether speakers present climate change as an urgent crisis or as an exaggerated issue?

Data Collection

Google

"Who are the most famous progressive/conservative YT political commentators?"

YouTube
Data API

Speaker name
+
"climate change"

+ quick manual check of the videos content

YouTube
Transcript API

Raw transcript of

34 videos

3 speakers
2 hrs, 30 minutes

11 speakers
1 hr, 20 minutes

Text Preprocessing (1)

PROBLEM	*SOLUTION*
The raw text transcript of each video is very long, and this might create some problems later.	Each transcript was split into chunks of max 250 words.
YouTube transcripts come in a raw version, without punctuation.	Used a punctuation restoration model, so that the models will be able to understand the context and meaning of the text better.
YouTube commantaries cover many topics, often unrelated to our target topic (climate change).	Used a semantic filtering system to filter out the text chunks which didn't cover topics related to my target.

Text Preprocessing (2)

RESTORING PUNCTUATION

Original text (example):

"[...] obama stopped paris parasailing is that the word i think that's the word parasailing with billionaires [...]"

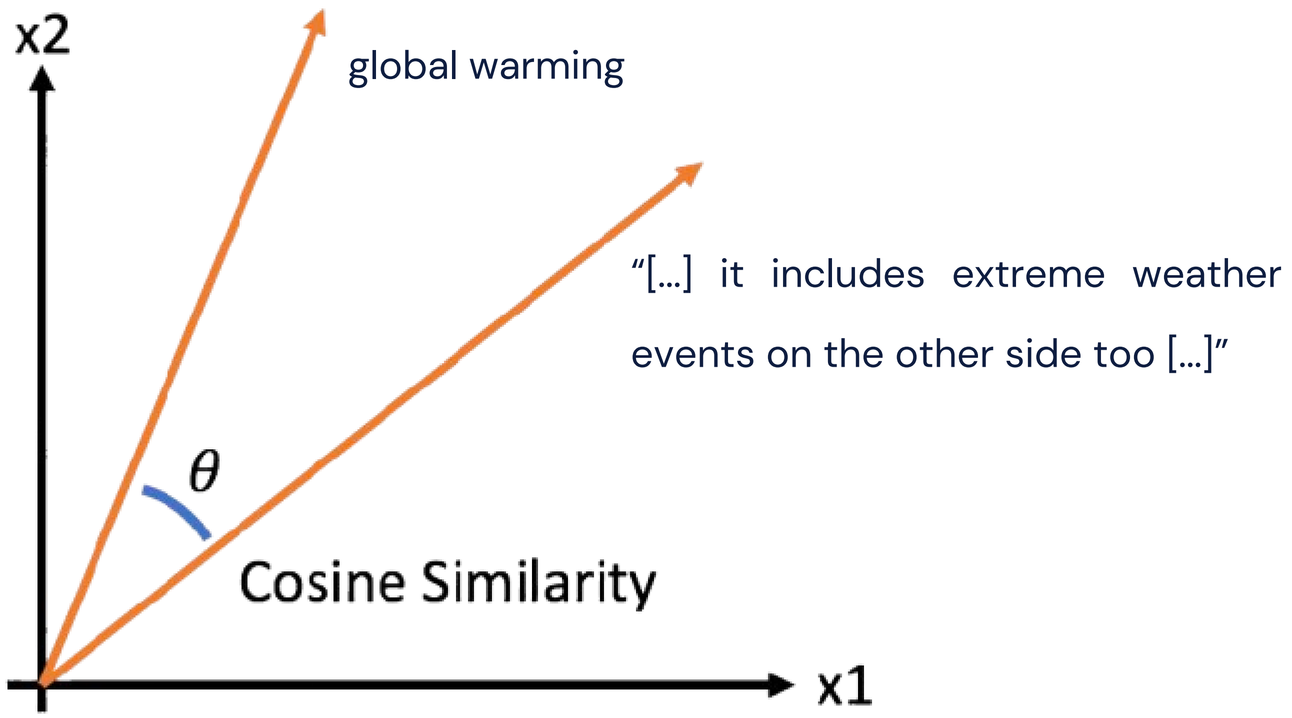
Text after punctuation restoration:

"[...] Obama stopped paris- parasailing- is that the word? i think that's the word. parasailing with billionaires [...]"

FILTERING OUT NON RELEVANT CHUNKS

Computed cosine similarity of the embeddings of each chunk with the following concepts, and kept only the chunks with at least 1 similarity greater than 0.4:

- climate change
- global warming
- carbon emissions
- green energy policies
- fossil fuel industry
- climate activism
- environmental regulation
- climate policy debate
- renewable energy transition
- carbon tax



Zero-shot Classification

To determine the stance of each speaker based on the chunks, we need LABELS.

We could use the label defined by the general ideology of each speaker ("progressive" vs "conservative"), but it wouldn't be very accurate.

A better solution is using

Zero-shot Classification

Hugging Face

Political_DEBATE_base_v1.0

A classifier trained for zero-shot and few-shot classification of political documents.

As we could expect, most of the "progressive" speakers believe it's an urgent crisis, and the opposite for the "conservative". This stregthens the power of our labels.

The following labels have been chosen:

Believes climate change is an urgent crisis

VS

Believes climate change is exaggerated

Stance Detection (1)

From cleaned text transcriptions

Hugging Face

all-MiniLM-L6-v2

To text embeddings

From having no labels

Hugging Face

Political_DEBATE_base_v1.0

To binary pseudo-labels

X

Y

LOGISTIC REGRESSION MODEL

With grid search *cross-validation*

	precision	recall	f1-score	support
False	0.76	0.94	0.84	17
True	0.80	0.44	0.57	9
accuracy			0.77	26
macro avg	0.78	0.69	0.71	26
weighted avg	0.78	0.77	0.75	26

After splitting the data into training and testing, the F1 Score for the chunks classified as skeptic is lower, since the absolute number of chunks with this label is lower compared to the other label.

Next, I retrained the model on the whole dataset, with the best parameters, to assign a new label to each chunk.

But why new labels?

Stance Detection (2)

ONLY USING ZERO-SHOT LABELS

Unstable

Not specific to our data

WEAK SUPERIVISION MODEL

More stable and consistent

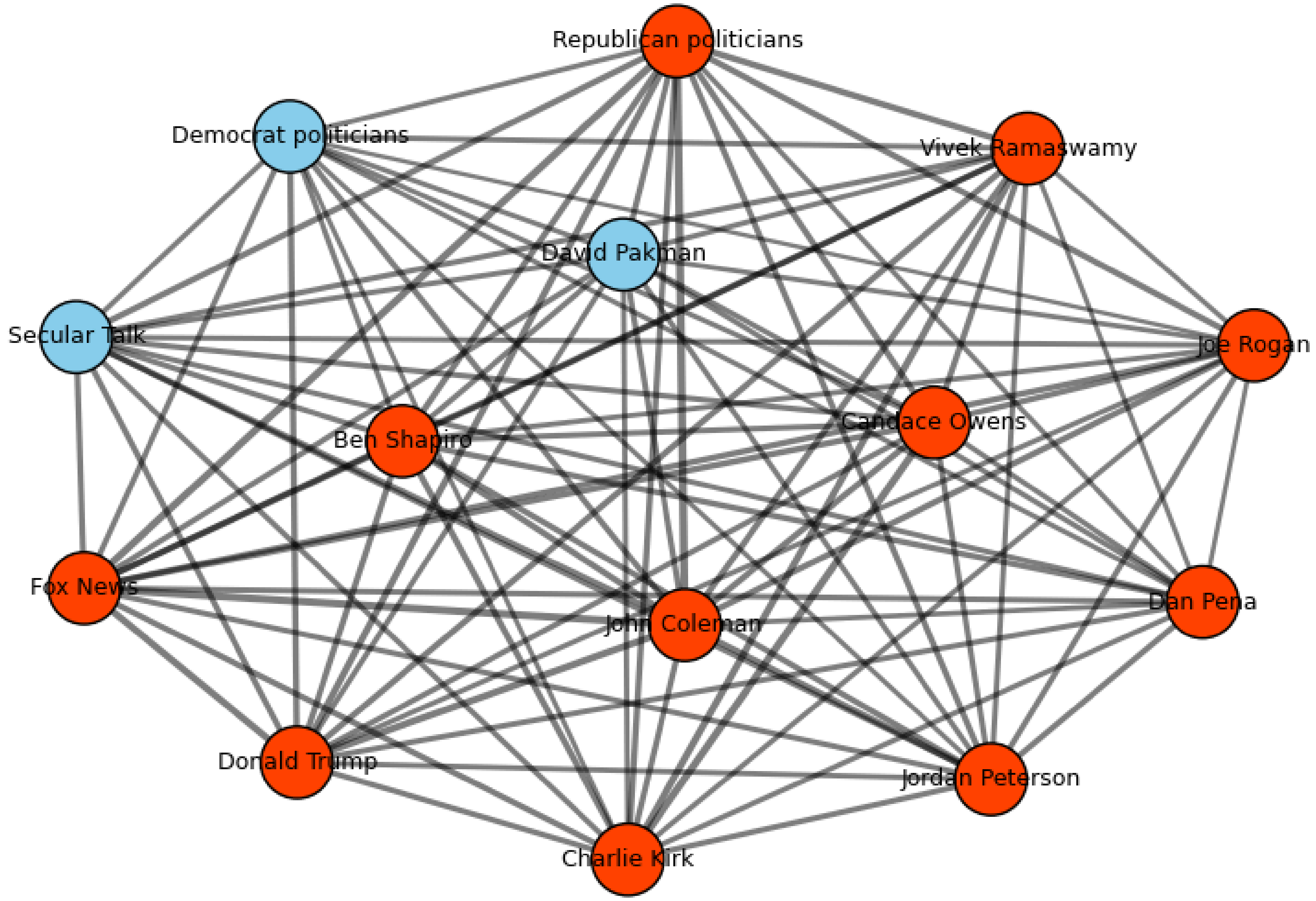
Adapted to our data

Can be improved with human-labeled data

Allows cross-validation and hyperparameter tuning

Since the logistic regression model is only exposed to the zero-shot model's outputs and not the true ground-truth labels, it cannot surpass the zero-shot model in terms of real accuracy.

Channels similarity



Calculated the average embedding of each channel

↓

Computed the cosine similarity between the mean vector of each channel

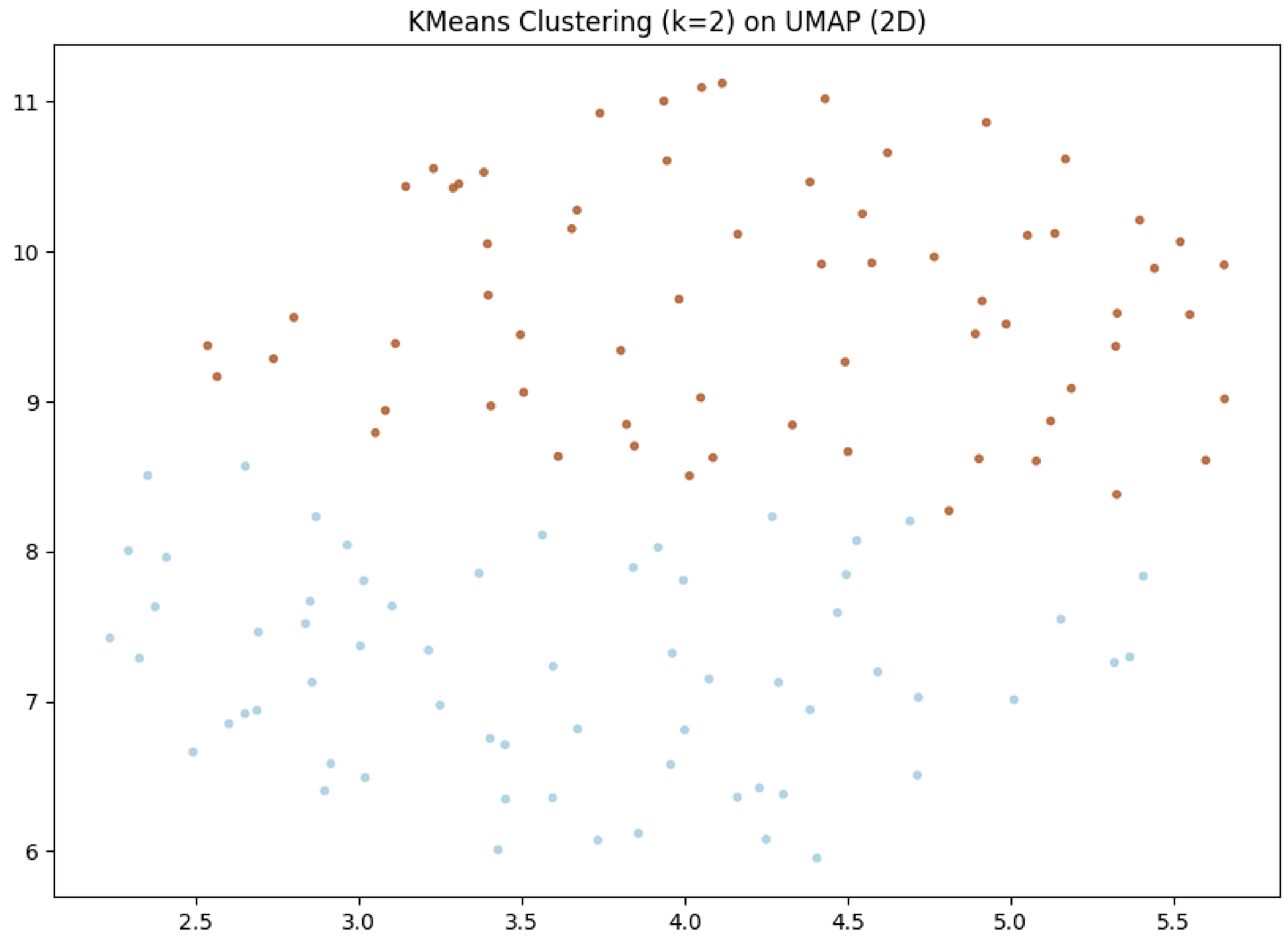
↓

Visualized using a *network graph*

Clustering

Dimensionality reduction using **HDBSCAN**

K-Means clustering with 2 clusters



Interpretation of the clustering results:

Label = 1

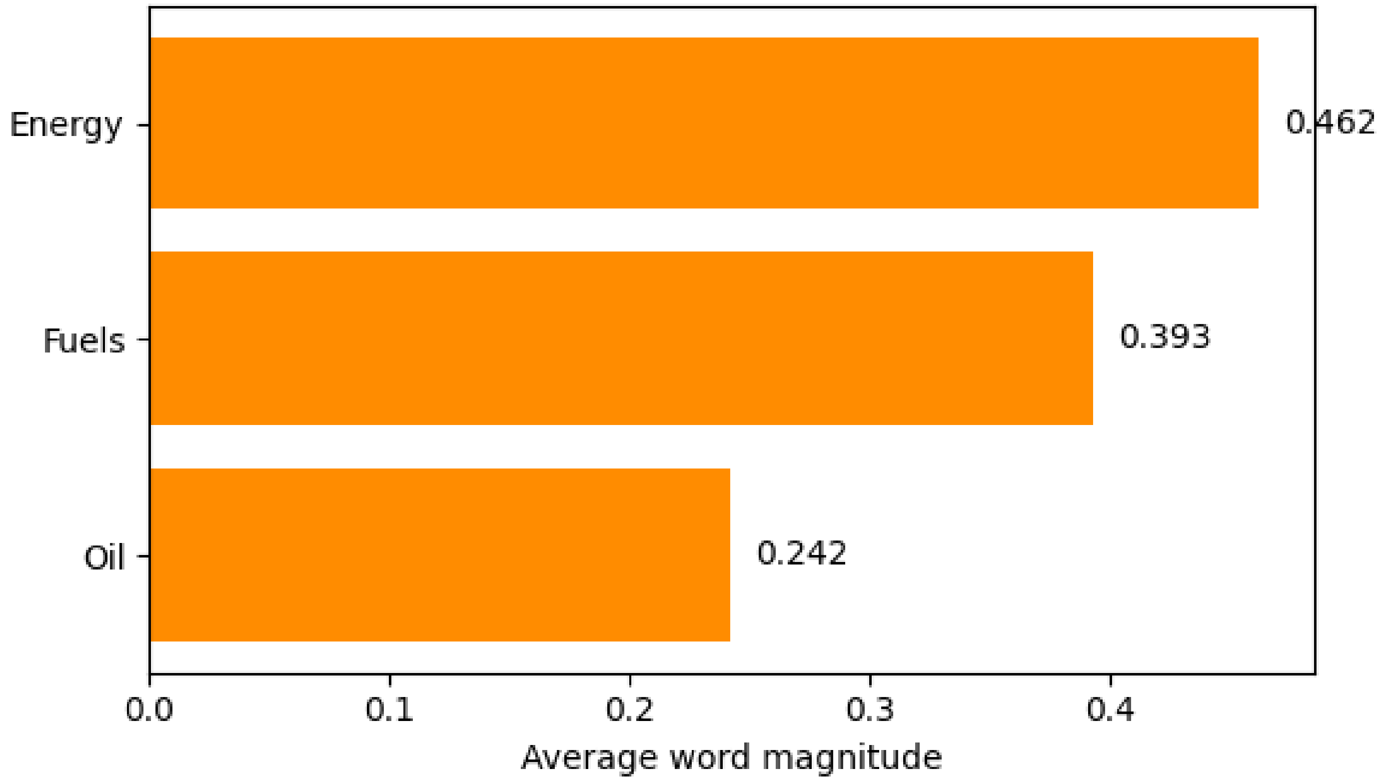
The speaker believes in a urgent crisis

Label = 0

The speaker believes climate change is an exaggeration

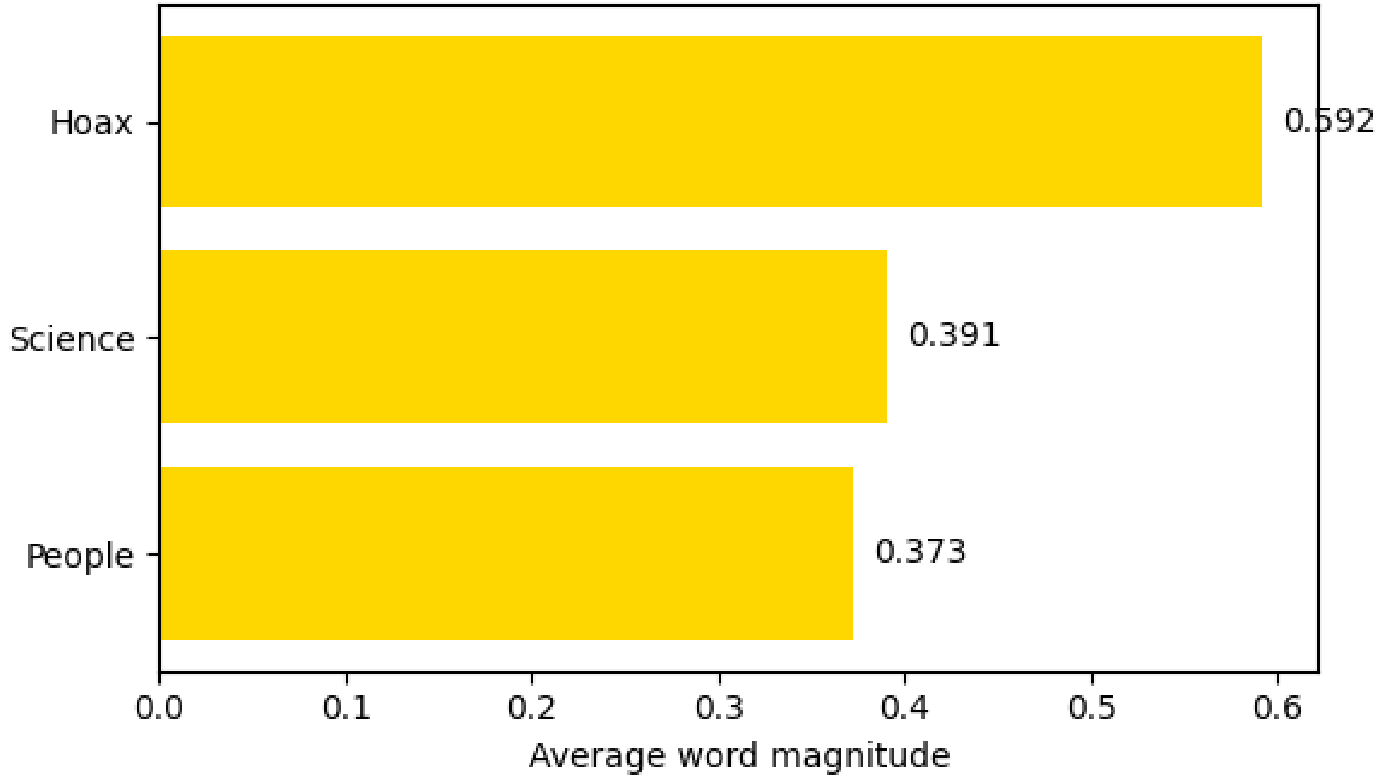
		count
lr_believes_exaggeration	kmeans_cluster	
False	1	57
	0	38
True	0	25
	1	8

Explainability with LIME



Chunks classified as "*believes climate change is an urgent crisis*" emphasize energy production, fossil fuels, and environmental impact, reflecting a material, policy-focused framing.

Chunks classified as “*believes climate change is exaggerated*” are driven by rhetorical markers of doubt and vague references to “science” or “people”, which reflect a more generalized, argumentative tone.



Extra

It is interesting to analyze the **toxicity** of the language of each speaker.

For this, I used the *Detoxify* model from GitHub.

Most toxic speakers (by average chunk toxicity):

Dan Pena —— **0.393**

Joe Rogan —— **0.325**

Fox News —— **0.312**

We can also use **named-entity recognition (NER)** to look at the most mentioned entities by the speakers.

For example, we can notice that in chunks that have been classified as "climate change is exaggerated", there are mentions of the Bible, of Al Gore and of the Dunning–Kruger effect.

On the other hand, the opposite chunks have multiple mentions of Big Oil, the Congress and the Senate.

Some of the people mentioned in the videos.

Final Remarks

Combining weak supervision (*zero-shot*) with interpretable models (*logistic regression* + *LIME*) yields robust and transparent insights.

Embeddings + *similarity analysis* reveal structural patterns in political communication

Political ideology shapes the framing of climate change on YouTube: progressive commentators emphasize urgency, while conservative voices focus more on uncertainty and counter-narratives.

NLP techniques successfully identify whether a speaker treats climate change as urgent or exaggerated, showing that stance is encoded in linguistic patterns.

THANK YOU

For your attention